CSE113
Foundations of Computer Science

Sequences

State University of New York, Korea

Some slides taken from Prof. Pramod Ganapathi (Stony Brook). Thanks!

"Mathematics is the art of giving the same name to different things."

- Henri Poincaré

What are sequences?

Types of sequences

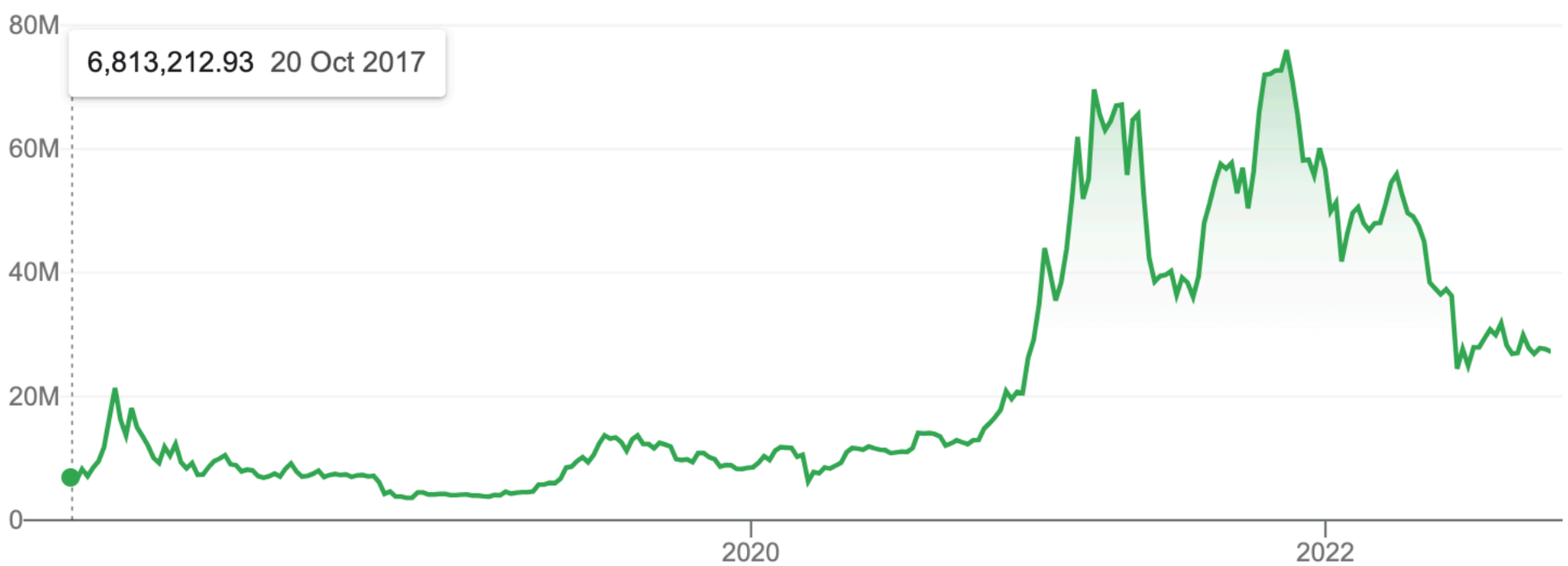
- Finite sequence: $a_m, a_{m+1}, a_{m+2}, \ldots, a_n$
  e.g.: $1^1, 2^2, 3^2, \ldots, 100^2$
- Infinite sequence: $a_m, a_{m+1}, a_{m+2}, \ldots$
  e.g.: $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots$

Term

- Closed-form formula: $a_k = f(k)$
  e.g.: $a_k = \frac{k}{k+1}$
- Recursive formula: $a_k = g(k, a_{k-1}, \ldots, a_{k-c})$
  e.g.: $a_k = a_{k-1} + (k-1)a_{k-2}$

Growth of sequences

- Increasing sequence
  e.g.: $2, 3, 5, 7, 11, 13, 17, \ldots$
- Decreasing sequence
  e.g.: $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots$



Sums and products of sequences

Sum

- Summation form:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n$$

where, $k =$ index, $m =$ lower limit, $n =$ upper limit
e.g.: $\sum_{k=m}^{n} \frac{(-1)^k}{k+1}$

Product

- Product form:

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \cdots \cdot a_n$$

where, $k =$ index, $m =$ lower limit, $n =$ upper limit
e.g.: $\prod_{k=m}^{n} \frac{k}{k+1}$

Properties of sums and products

- Suppose $a_m, a_{m+1}, a_{m+2}, \ldots$ and $b_m, b_{m+1}, b_{m+2}, \ldots$ are sequences of real numbers and c is any real number

Sum

- $\sum_{k=m}^{n} a_k = \sum_{k=m}^{i} a_k + \sum_{k=i+1}^{n} a_k$ for $m \leq i < n$ where, i is between m and $n-1$ (inclusive)
- $c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} (c \cdot a_k)$
- $\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k)$

Product

- $(\prod_{k=m}^{n} a_k) \cdot (\prod_{k=m}^{n} b_k) = \prod_{k=m}^{n} (a_k \cdot b_k)$

Proof by Mathematical Induction

Proof for dominos

Core idea

- A starting domino falls. From the starting domino, every successive domino falls. Then, every domino from the starting domino falls.

Mathematical Induction

- Let P(n) denote a predicate, and our objective is to prove P(n) for all integers n>=0. Mathematical induction works in two steps.
- Base step: prove P(0)
- Inductive step: prove P(k) -> P(k+1) for any k>=0

Proof by mathematical induction: Example 1

Proposition

- $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{n}\right)=\frac{1}{n}$ for all integers $n \geq 2$.

Proof by mathematical induction: Example 1

Proposition

- $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{n}\right)=\frac{1}{n}$ for all integers $n \geq 2$.

Proof

Let $P(n)$ denote $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{n}\right)=\frac{1}{n}$.

- Basis step. $P(2)$ is true. $\triangleright$ How?
- Induction step.

  Assume $P(k)$: $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{k}\right)=\frac{1}{k}$ for some $k \geq 2$.

  Prove $P(k+1)$: $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{k+1}\right)=\frac{1}{k+1}$

  LHS of $P(k+1)$

  $=\left[\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{k}\right)\right]\left(1-\frac{1}{k+1}\right)$

  $=\frac{1}{k}\left(1-\frac{1}{k+1}\right)$ $\quad(\because P(k)$ is true$)$

  $=\frac{1}{k}\cdot\frac{k}{k+1}$ $\quad(\because$ common denominator$)$

  $=\frac{1}{k+1}$ $\quad(\because$ remove common factor$)$

  $=$ RHS of $P(k+1)$

Proof by mathematical induction: Example 2

Proposition

- Fibonacci sequence is: $F(0) = 1$, $F(1) = 1$, and $F(n) = F(n-1) + F(n-2)$ for $n \geq 2$. Prove that: $F(0)^2 + F(1)^2 + \cdots + F(n)^2 = F(n)F(n+1)$ for all $n \geq 0$.

Proof by Mathematical Induction — Guideline

- The problem usually looks like this:
  - Prove: For any integer n >=1, P(n) holds
- Check P(1), P(2) … before the proof
- Try to solve P(k) -> P(k+1)
- Write a proof that has:
  - A notation for predicate P(n)
  - A base step
  - An inductive step

Exercises

Use mathematical induction to prove the following identities. **Final 2021**

(a) [5 points] For all integers $n \geq 1$,

$$\sum_{i=1}^{n} i(i!) = (n+1)! - 1.$$

Use mathematical induction to prove the following identities.

(a) [5 points] For all integers $n \geq 1$,

$$\sum_{i=1}^{n} i(i!) = (n+1)! - 1.$$

- **Proof**.
  - We will prove this proposition with mathematical induction.
  - **Let P(n) denote the predicate** 1(1!) + 2(2!) + … + n(n!) = (n+1)! -1
  - **Base step**: **We prove** P(1).
    - LHS is 1, and RHS is 2! - 1=1
  - **Inductive step: We prove** P(k) -> P(k+1) for all integer k>=1
    - Let k be an arbitrary integer and k>=1.
    - Assume P(k) holds. That is 1(1!) + 2(2!) + … + k(k!) = (k+1)! -1
    - We want to prove P(k+1), namely, 1(1!) + 2(2!) + … + k(k!) + (k+1) (k+1)! = (k+2)! -1
      - LHS can be reduced to (k+1)! -1 + (k+1) (k+1)! following assumption P(k)
      - The latter can be further reduced to (k+2)! -1, namely RHS.
- **QED**.

Use mathematical induction to prove the following identities.

(b) [5 points] Consider the Fibonacci sequence: $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$, for $n \geq 2$. For all integers $n \geq 2$,

$$f_{n+2} = 1 + \sum_{i=0}^{n} f_i.$$

Use mathematical induction to prove the following identities.

(b) [5 points] Consider the Fibonacci sequence: $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$, for $n \geq 2$. For all integers $n \geq 2$,

$$f_{n+2} = 1 + \sum_{i=0}^{n} f_i.$$

- Proof.
  - We will prove this proposition with mathematical induction.
  - Let P(n) be the predicate f_{n+2} = 1+ f_0+f_1 +… + f_n
  - Base step: We first prove P(2) holds.
    - LHS = f_4 = 3. RHS = 1+ f_0 + f_1 + f_2 = 1+0+1+1=3
  - Inductive step:: Then we prove P(k) -> P(k+1) for all k>=2
    - Let k be an arbitrary integer and k>=2
    - Assume P(k) holds, namely, f_{k+2} = 1+ f_0+f_1 +… + f_k
    - We want to prove P(k+1), that is, f_{k+3} = 1+ f_0+f_1 +… + f_k+f_{k+1}
      - LHS = f_{k+2}+f_{k+1} from Definition of Fobonacci Sequence.
      - Thus, LHS = 1+ f_0+f_1 +… + f_k + f_{k+1} following assumption P(k)
      - Thus equals RHS
- QED.=

Final 2020

Use mathematical induction to prove the following identities.

(a) [5 points] For all natural numbers n,

$$1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + n^2 \times (n+1) = \frac{n(n+1)(n+2)(3n+1)}{12}$$

Final 2020

Use mathematical induction to prove the following identities.

(a) [5 points] For all natural numbers n,

$$1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + n^2 \times (n+1) = \frac{n(n+1)(n+2)(3n+1)}{12}$$

- Proof.
  - We will prove this proposition with mathematical induction.
  - Let P(n) be the predicate 1^2 * 2 + … + n^2 * (n+1) = n (n+1) (n+2) (3n+1)/12
  - Base step: We first prove P(1) holds.
    - LHS = 1^2 * 2=2. RHS = 1*2*3*4/12 = 2.
  - Inductive step: Then, we prove P(k) -> P(k+1) for all k>=1
    - Let k be an arbitrary integer and k>=1.
    - Assume P(k) holds. That is, 1^2 * 2 + … + k^2 * (k+1) = k (k+1) (k+2) (3k+1)/12
    - We want to prove P(k+1), that is, 1^2 * 2 + … + k^2 * (k+1) + (k+1)^2 * (k+2) = (k+1) (k+2) (k+3) (3k+4)/12
      - LHS = k (k+1) (k+2) (3k+1)/12 + + (k+1)^2 * (k+2) following assumption P(k)
      - The latter can be further reduced to (k+1)(k+2)/12 * (3k^2+k + 12k+12) = RHS
- QED.

Use mathematical induction to prove the following identities.

(b) [5 points] For all natural numbers n,

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n \times (n+1)} = \frac{n}{n+1}$$

Final 2020

Use mathematical induction to prove the following identities.

(b) [5 points] For all natural numbers n,

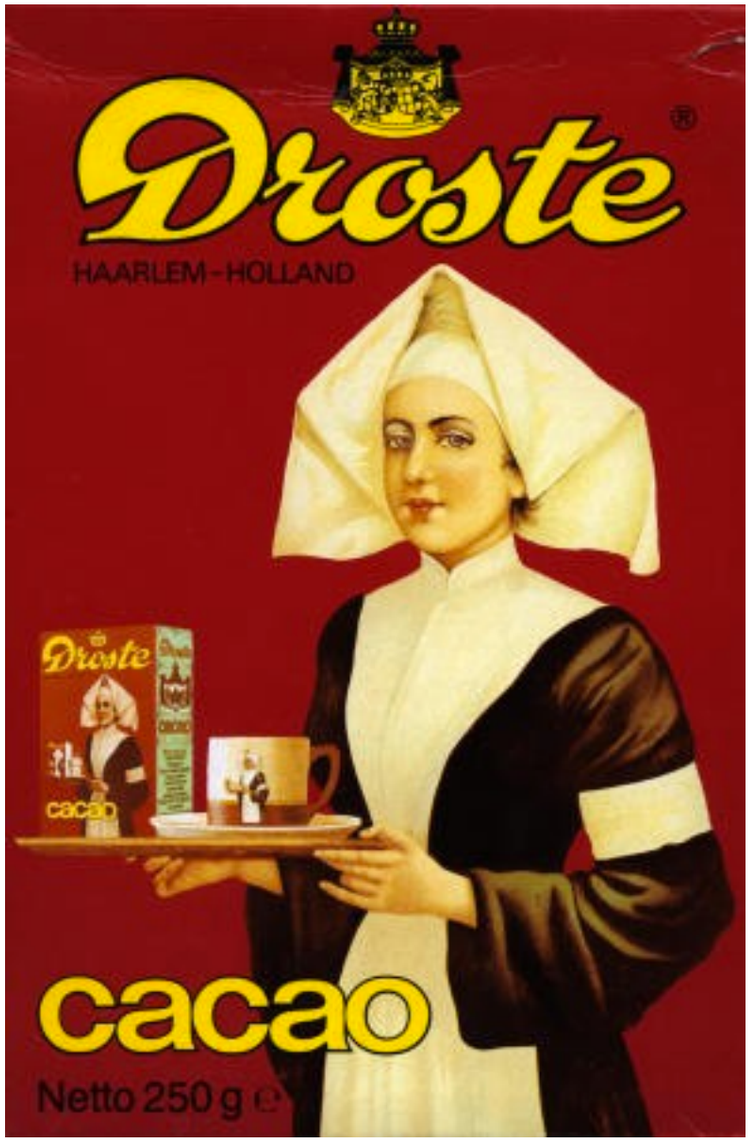
$$\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{n\times(n+1)}=\frac{n}{n+1}$$

- Proof.
  - We will prove this proposition with mathematical induction.
  - Let P(n) be the predicate 1/(1*2) + 1(2*3) + … + 1/(n *(n+1)) = n/(n+1)
  - Base step: We first prove P(1) holds.
    - LHS is 1/2, and RHS is 1/(1+1) = 1/2
  - Inductive step: Then, we prove P(k) -> P(k+1) for all integer k>=1
    - Let k be an arbitrary integer and k>=1.
    - Assume P(k) holds. That is 1/(1*2) + 1(2*3) + … + 1/(k*(k+1)) = k/(k+1)
    - We want to prove P(k+1), namely,1/(1*2) + 1(2*3) + … + 1/(k*(k+1)) + 1/((k+1)*(k+2))= (k+1)/(k+2)
      - LHS can be reduced to k/(k+1) + 1/((k+1)*(k+2)) following the assumption P(k)
      - The latter can be further reduced to (k^2+2k+1)/((k+2)(k+1)), which equals to RHS.
- QED.

Recursive sequences

Recursion = Repeating itself

- recursive sequences
- recursive functions
- recursive data structures



Example

Examples

- Suppose $f(n) = n!$, where $n \in \mathbb{W}$. Then,

$$f(n) = \begin{cases} 1 & \text{if } \ n = 0, \\ n \cdot f(n-1) & \text{if } \ n \geq 1. \end{cases}$$

Closed-form formula: $f(n) = n \cdot (n-1) \cdot \cdots \cdot 1$

Examples

Examples

- Suppose $f(n) = n!$, where $n \in \mathbb{W}$. Then,
  $$f(n) = \begin{cases} 1 & \text{if } \ n = 0, \\ n \cdot f(n-1) & \text{if } \ n \geq 1. \end{cases}$$
  Closed-form formula: $f(n) = n \cdot (n-1) \cdot \cdots \cdot 1$
- Suppose $F(n) = n$th Fibonacci number. Then,
  $$F(n) = \begin{cases} 1 & \text{if } \ n = 0 \text{ or } 1, \\ F(n-1) + F(n-2) & \text{if } \ n \geq 2. \end{cases}$$
  Closed-form formula: $F(n) = ?$

Example： Arithmetic sequence

Let $a_0, a_1, a_2, \ldots$ be the sequence defined recursively as follows: For all integers $k \geq 1$,

(1) $a_k = a_{k-1} + 2$ recurrence relation

(2) $a_0 = 1$ initial condition.

- Write out a_1, a_2, a_3, a_4, and a_5
- Derive an explicit formula of the sequence
- Confirm the explicit formula satisfies the recursive definition

Example: Geometric sequence

You deposit $100,000 in a bank account for a 3% interest compounded annually. How much will you get after 21 years?

Solution

- Suppose $A_k =$ Amount in your account after k years. Then,

$$A_k = \begin{cases} 100,000 & \text{if } \ k = 0, \\ (1 + 3\%) \times A_{k-1} & \text{if } \ k \geq 1. \end{cases}$$

- Solving the recurrence by the method of iteration, we get

$$\boxed{A_k = ((1.03)^k \cdot 100,000) \text{ dollars}}$$

- Homework:

Confirm the explicit formula satisfies the recursive definition

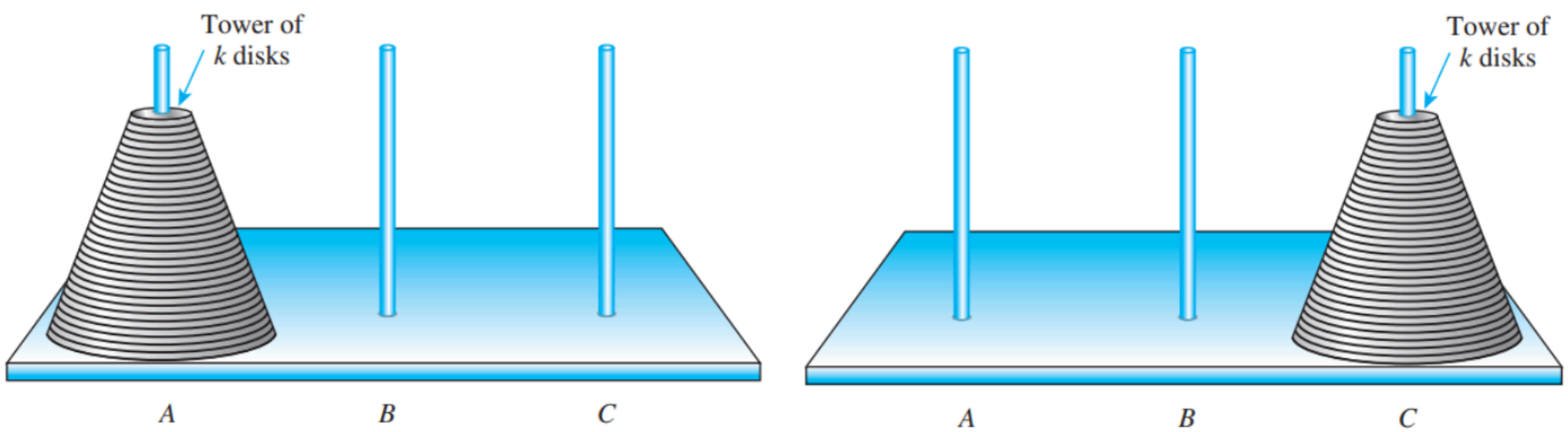
- $A_{21} = ((1.03)^{21} \cdot 100,000) \approx 186,029.46$ dollars

Application

Example: Towers of Hanoi

Problem

- There are k disks on peg 1. Your aim is to move all k disks from peg 1 to peg 3 with the minimum number of moves. You can use peg 2 as an auxiliary peg. The constraint of the puzzle is that at any time, you cannot place a larger disk on a smaller disk.
  What is the minimum number of moves required to transfer all k disks from peg 1 to peg 3?



Demo: https://www.mathsisfun.com/games/towerofhanoi.html

Example: Towers of Hanoi

Solution

Suppose $k = 1$. Then, the 1-step solution is:

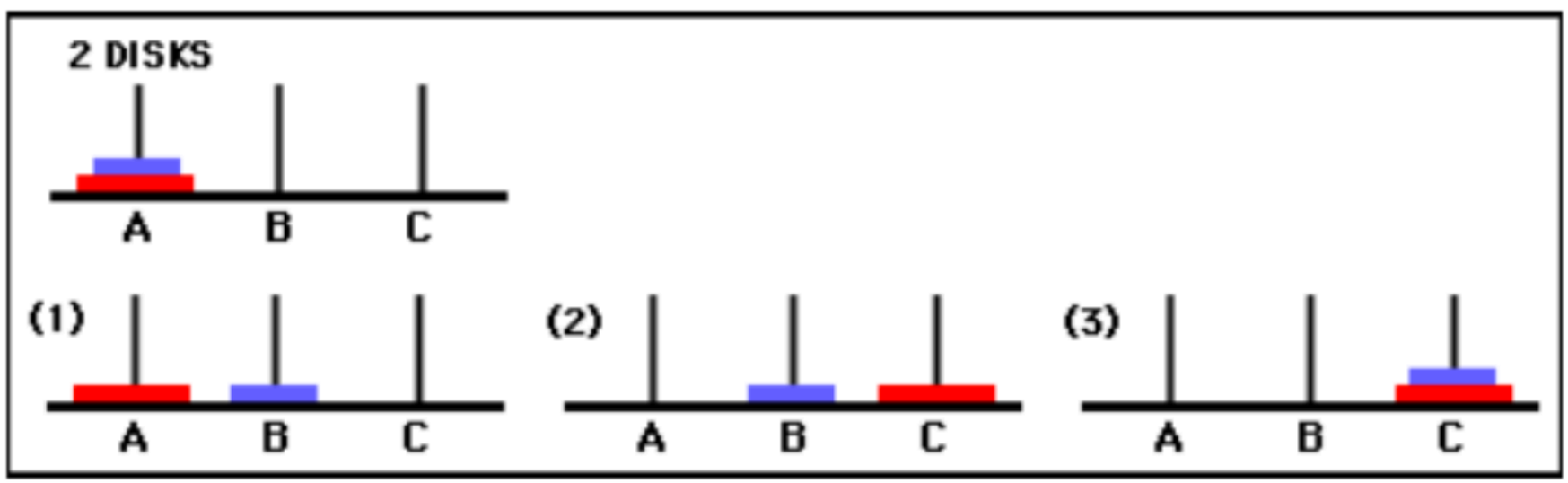
1. Move disk 1 from peg A to peg C.

Example: Towers of Hanoi

Solution

Suppose $k = 2$. Then, the 3-step solution is:

1. Move disk 1 from peg A to peg B.
2. Move disk 2 from peg A to peg C.
3. Move disk 1 from peg B to peg C.

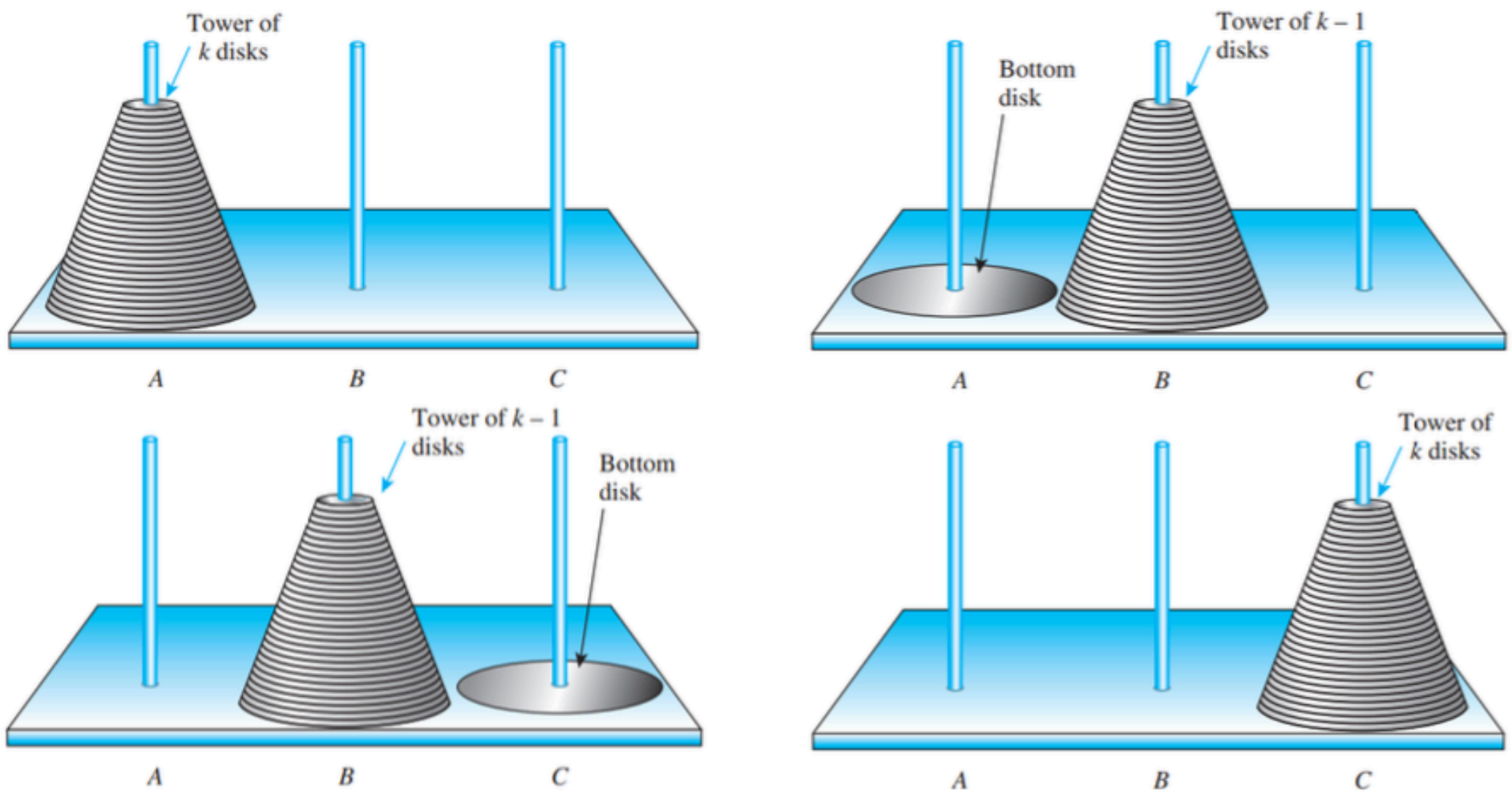


Example: Towers of Hanoi

Solution

For any $k \geq 2$, the recursive solution is:

1. Transfer the top $k-1$ disks from peg A to peg B.
2. Move the bottom disk from peg A to peg C.
3. Transfer the top $k-1$ disks from peg B to peg C.

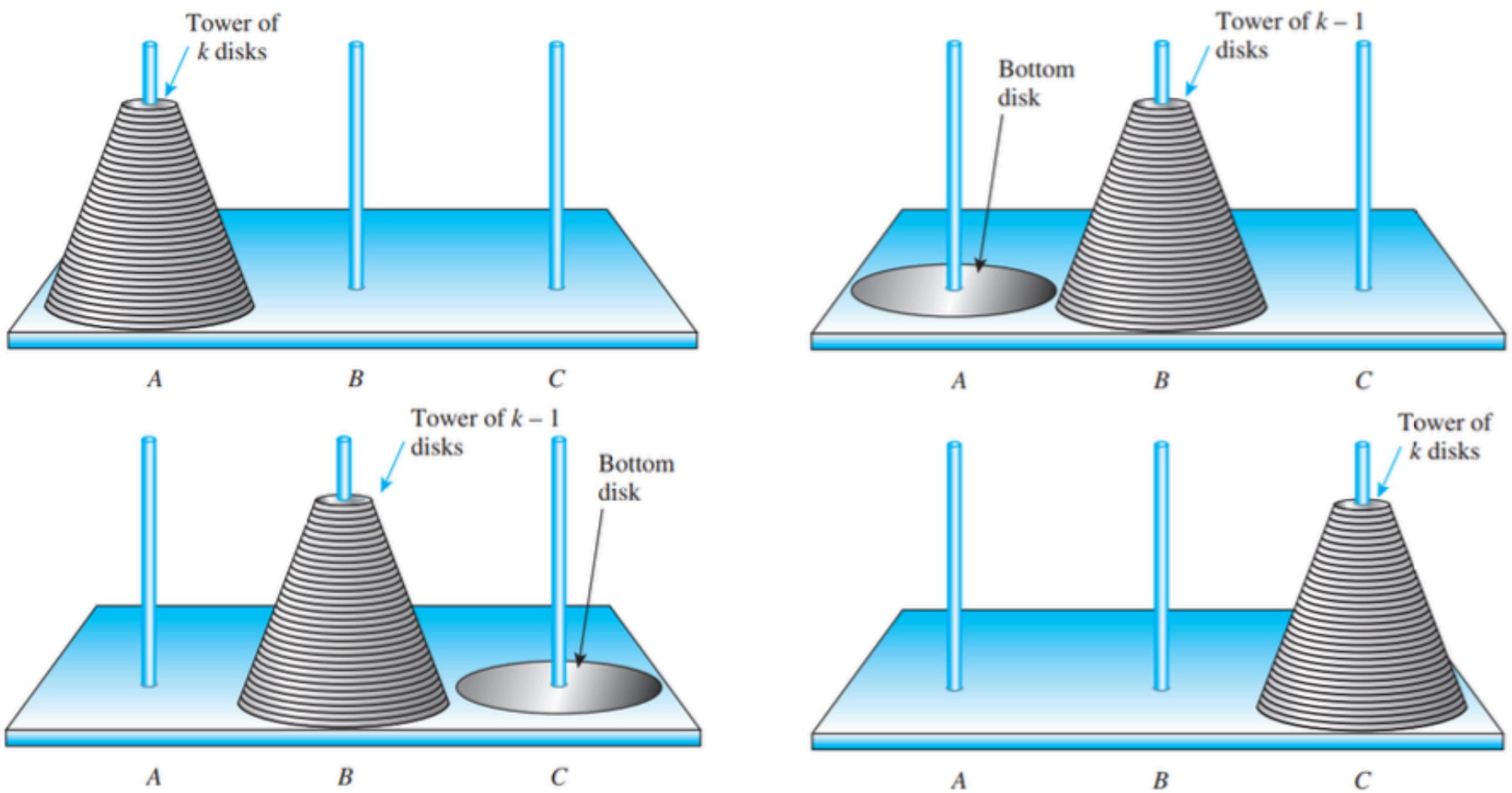


Example: Towers of Hanoi

Code Demo

TOWERS-OF-HANOI(k, A, C, B)

1. if $k = 1$ then
2. Move disk k from A to C.
3. elseif $k \geq 2$ then
4. TOWERS-OF-HANOI$(k-1, A, B, C)$
5. Move disk k from A to C.
6. TOWERS-OF-HANOI$(k-1, B, C, A)$



Example: Towers of Hanoi

Solution (continued)

- Let $M(k)$ denote the minimum number of moves required to move k disks from one peg to another peg. Then
$$M(k) = \begin{cases} 1 & \text{if } k = 1, \\ 2 \cdot M(k-1) + 1 & \text{if } k \geq 2. \end{cases}$$
- Solving the recurrence by the method of iteration, we get
$$\boxed{M(k) = 2^k - 1}$$
▷ How?

Why minimum?
https://proofwiki.org/wiki/Tower_of_Hanoi

Exercises

Exercise 1

- Consider the recursive sequence below. Find its explicit form and prove your answer, namely Confirm the explicit formula satisfies the recursive definition

$$a_k = ka_{k-1}, \text{ for all integers } k \geq 1$$
$$a_0 = 1$$

Solution

We have: $a_k = k \cdot a_{k-1} = k \cdot (k-1) \cdot a_{k-2} = k(k-1)(k-2) \cdots 1 \cdot a_0 = k!$

- To prove it indeed satisfies the recursive definition, we plug it to the recursive definition:
- The explicit form clearly satisfies a_0=1
- The explicit form also satisfies a_k = k a_{k-1}, since the left is k!, the right is k * (k-1)! which is also k!

Exercise 2

- Consider the recursively defined sequence below. Find its explicit form and prove your answer.

$$b_k = \frac{b_{k-1}}{1 + b_{k-1}}, \text{ for all integers } k \geq 1$$
$$b_0 = 1$$

Solution

- We have $b_0=1$, $b_1= 1/2$, $b_2=1/3$, $b3=1/4$… So a possible explicit form is $b_n=1/(n+1)$
- To prove it indeed satisfies the recursive definition, we plug it to the recursive definition:
  - The explicit form clearly satisfies $b_0=1$
  - The explicit form also satisfies $b_k = b_{k-1} / (1+b_{k-1})$, since LHS=$1/(k+1)$, RHS = $1/(k+1) / (1 + 1/(k+1)) = 1/(k+1)$

Exercise 3

- Consider the recursively defined sequence below. Find its explicit form and prove your answer.

$$c_k = 3c_{k-1} + 1, \text{ for all integers } k \geq 2$$
$$c_1 = 1$$

Solution

- We have c_1=1, c_2= 4, c_3=13, b4=40… So a possible explicit form is c_n=(3^n-1)/2
- To prove it indeed satisfies the recursive definition, we plug it to the recursive definition:
  - The explicit form clearly satisfies c_1=1
  - The explicit form also satisfies_____, since LHS= ___ RHS = ___ = ____